MAKKINA KARTHIK

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Education

•Bachelor of Technology in Computer Science and Engineering *2022 - 2026*
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi CGPA: 8.3

Technical Skills and Interests

Languages: Python, MySQL, C
DevOps Cloud: : Linux (Ubuntu), Microsoft Azure, Jenkins, CI/CD, Docker
Version Control: Git, GitHub
Data Analysis: Pandas, NumPy, Data Cleaning, DAX
Visualization: Power BI

Projects

•Leaf disease detection using Machine Learning
Technologies used : Python, Google colab, Image Processing

- Engineered an image-based plant disease detection system that computes spectral vegetation indices (NDVI, EVI, SAVI, etc.) using OpenCV and NumPy from leaf images stored in Google Drive.
- Preprocessed 5,000+ image samples through normalization and feature extraction to improve model quality.
- Deployed the solution on Google Colab, integrating Google Drive for dataset access and Adafruit-like CSV streaming for result analysis.

•Chocolate Industry Sales Analytics Dashboard using Power BI
Technologies used: Power BI, DAX, Power Query, Excel, Data Modeling

- Designed an interactive Power BI dashboard analyzing 10,000+ sales transactions across multiple regions.
- Built 15+ DAX measures to calculate KPIs including revenue, profit margin, and sales growth. Reduced manual reporting effort by 60% using automated dashboards.
- Improved executive decision-making through drill-down reports and dynamic filters. Developed star schema data model improving dashboard performance by 35%.

•Cold Email Generator using LLaMA 3.1 (via Groq API)
Technologies used: Python, LangChain, Groq API (LLaMA 3.1), ChromaDB, Streamlit

- Built an AI-powered Cold Email Generator using LLaMA 3.1 (via Groq API) and Python, which parses job descriptions to automatically extract role, required skills, and experience.
- Developed an information extraction pipeline achieving 95% + accurate extraction of job skills from descriptions. Automated portfolio matching, reducing manual email drafting time by approximately 90%.
- Applied LLM prompt-engineering and API integration skills (Groq, LLaMA 3.1) to automate a real-world outreach workflow, showcasing practical experience in applied AI/LLM development — relevant to both DevOps automation and data-driven engineering roles.

INTERNSHIPS

•UI/UX Designer Intern | Greensort *Sep 2024 - Nov 2024*
Tools - Figma, Microsoft Designer, ChatGPT, Canva Offline

- Designed website to improve the user to navigate the website in an easy manner and reached 2000+ visitors within a week after launching the website.

•UI/UX Designer Intern | EazyByts *Jan 2025*
Tools - Figma, ChatGPT, Freepik, Microsoft Copilot Online

- Worked on designing various projects such as an event management app, digital marketing agency website, stocks trading app designs which improved my design thinking, time managing and understanding the designs from user perspective.

CERTIFICATIONS

Python Programming Fundamentals Infosys Springboard
TalentNext Course on Database Solutions Wipro TalentNext